



NEW PRODUCTS

MAKE IN INDIA

COMPONENTS

Optical heart rate sensor

ROHM has introduced an optical heart rate sensor for smart bands and watches that is capable of measuring blood pressure, stress and vascular age at a high-speed 1024Hz sampling rate. The BH1792GLC consumes only 0.44mA (during heart rate measurement), contributing to longer operating time. 1024Hz sampling rate makes it possible to measure heart rate up to 32 times faster than conventional solutions while also supporting vital sign sensing such as vascular age and stress measurement demanded by the latest wearable and fitness devices.

ROHM Semiconductor
www.rohm.com

System-on-chip for autonomous vehicles

Renesas Electronics has announced R-Car V3H system-on-chip (SoC) that delivers high computer-vision performance and artificial intelligence (AI) processing at extremely low power levels for automotive front cameras in mass-produced Level 3 (conditional automation) and Level 4 (high automation) autonomous vehicles. The new chip is optimised for use in stereo front cameras and achieves five times the computer vision performance of its predecessor. Part of the open Renesas autonomy platform for advanced driver-assistance systems and automated driving, it enables design flexibility for Tier 1s and OEMs to map their own road from assisted to highly automated vehicles.

Renesas Electronics
www.renesas.com

GaN drivers

Expanding its gallium-nitride (GaN) power portfolio, Texas Instruments has

added two new high-speed GaN field-effect transistor (FET) drivers for speed-critical applications such as light detection and ranging (LIDAR) and 5G radio-frequency (RF) envelope tracking. The LMG1020 and LMG1210 can deliver 50MHz switching frequency while improving efficiency and enabling five times smaller solution sizes than those possible with silicon MOSFETs.

Texas Instruments
www.ti.com

Power modules

STMicroelectronics' new ACEPACK (Adaptable Compact Easier PACK-age) modules provide cost-effective and highly integrated power conversion for 3-30kW applications

including industrial motor drives, air-conditioners, solar generators, welders, battery chargers, UPS controllers and electric vehicles. Features include optional solder-free press-fit connections, which simplify assembly as an alternative to conventional soldered pins, and metal screw clamps that enable fast and reliable mounting.

Designers have the choice of six-pack modules that contain six IGBTs with freewheel diodes as a three-phase inverter, and power integrated modules that provide a complete driver power stage.

STMicroelectronics
www.st.com/acepack

SiC MOSFETs

Littelfuse has added two 1200V silicon carbide (SiC) n-channel, enhancement-mode MOSFETs to its first-generation portfolio of power semiconductor devices.

Developed in partnership with Monolith, these MOSFETs are designed to outperform silicon MOSFETs and IGBTs for ultra-fast switching in power conversion systems.

Littelfuse, Inc.
www.littelfuse.com

LEDs

Everlight Electronics has developed 5630X-ELB LED series that not only provides 228lm/W (at 65mA, 5000K, CRI > 80) but also meets DLC4.0 with light-on tests in high temperature range



(105°C), L90 > 36,000 hours and colour tolerance less than 3SDCM. Thus, these LEDs combine the advantages of luminous efficacy and stability for commercial, professional and industrial applications.

Everlight Electronics
www.everlight.com

BLDC fan driver IC

Allegro MicroSystems has introduced a new sensorless BLDC motor controller that is claimed to be the most advanced solution on the market today. Automotive version of A5932, it is a three-phase solution for high-power automotive fan applications with very low vibration and audible noise requirements. MOSFET gate driver outputs permit motor currents that can be scaled to fit the application requirements. User-friendly GUI interface allows fast evaluation and a shorter time to market. Key automotive applications include seat, battery, infotainment and headlamp cooling fans.

Allegro MicroSystems
www.allegromicro.com

Sensor for in-car cameras

ON Semiconductor is now offering an AEC-Q100 qualified image sensor

